

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: STATISTICS I (3.1)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Is Thika Road really well served? You have collected and studied data from the 8,000 m stretch of Thika Road near Ruaraka Hills Secondary School (the 20-gap sample). Answer each section in the space below it. Use numbers from our class data and always give the unit (metres, m). Write in full sentences. Where a question asks for a calculation, show your working. Remember: Ms Halima Abdi is not careless, so describe her claim as true or incomplete, not as wrong. Use the data table below for Section 2.

Grouped gap data, Thika Road, 20-gap sample:

Class (m)	Frequency
0 to <200	5
200 to <400	9
400 to <600	0
600 to <800	2
800 to <1000	1
1000 to <1200	3
Total	20

1. Checking the officer's claim

Ms Halima Abdi, the County Road Safety Officer, said: 'The average distance between crossing points is about 400 m, so the road is well served.' The

- (a) 21 designated crossing points make 20 gaps. Total length divided by number of gaps: $8,000 \text{ m} / 20 = 400 \text{ m}$. The mean gap is 400 m.
- (b) Her claim is true, because the mean really is 400 m. But it is incomplete. For the 20-gap sample on Thika Road the median gap is 250 m and the mode is 200 m, both much lower than 400 m. Fourteen of the 20 gaps are 300 m or shorter. A few very long gaps pull the mean up, so the one number does not show how the gaps are really spread.

stretch of Thika Road is 8,000 m long with 21 designated crossing points. (a) How many gaps are there, and how can you check her figure of 400 m? (b) Is her claim true? Is it complete? Explain using the ungrouped median and mode of the 20 gaps, which you worked out in class.	
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2. Reading the grouped table	
Use the grouped table in the instructions. (a) Which is the modal class? (b) How many gaps are shorter than 400 m, and how many are 600 m or longer? (c) Calculate the estimated mean gap using class midpoints. Show your working and say why it is called an estimate. (d) The class 400 to <600 is empty, yet your estimated mean from (c) is above 400 m. What does this tell you about the mean?	(a) The modal class is 200 to <400, with the highest frequency (9 gaps). (b) Shorter than 400 m: $5 + 9 = 14$ gaps. 600 m or longer: $2 + 1 + 3 = 6$ gaps. (c) Midpoints are 100, 300, 500, 700, 900, 1100. $f \times \text{midpoint}$: 500, 2700, 0, 1400, 900, 3300. Sum = 8,800. Estimated mean = $8,800 / 20 = 440$ m. It is an estimate because grouping hides the exact values, so we use the midpoint of each class in place of the real gaps. It differs from the true mean of 400 m. (d) The mean can fall in a place where there are almost no actual values. It is pulled up by the six long gaps, so it does not describe a typical gap on Thika Road (20-gap sample).

3. Using graphs to see the shape	
(a) Describe what a histogram of the equal-width table would look like. Mention where the tallest bar is, the empty class and the bars on the right. (b) When classes 400 to <600, 600 to <800 and 800 to <1000 are merged into 400 to <1000, why must we use frequency density instead of frequency for	(a) The tallest bar is at 200 to <400 (9 gaps). There is a tall bar at 0 to <200 (5). The bar at 400 to <600 is empty (a gap in the graph). Then there are small bars at 600 to <800 (2) and 800 to <1000 (1), and a separate bar of 3 at 1000 to <1200. Most gaps are short, with a few long ones to the right. (b) The merged class is three times wider than the others, so using frequency as height would make it look misleadingly tall. In a histogram, area shows frequency, so we use frequency density = frequency / class width. The merged class has 3 gaps (600, 700, 800) and width 600 m = 6 units of 100 m, so density = $3 / 6 = 0.5$. It is the lowest bar even though it holds 3 gaps. (c) The Mombasa Road polygon (frequencies 4, 8, 5, 2, 1, 0) is a single hump with a short tail and no gap of 1000 m or more. The Thika Road polygon (5, 9, 0, 2, 1, 3) has a second small rise at 1000 to <1200 m. So the means are similar (440 m and 380 m) but the shapes and spread differ. A mean alone cannot show this.

bar height? Calculate the frequency density of this merged class (use class width in units of 100 m). (c) Mombasa Road (Embakasi) has a grouped mean of 380 m and Thika Road 440 m. Using their frequency polygons, explain why similar means can hide different stories.	
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4. Where is the road really unsafe?	
Use the pedestrian and near-miss data from the class project: the long stretch G9-G14 (six gaps from 600 m to 1,100 m) includes the school gate. The whole 20-gap sample had 182 pedestrians crossing at undesignated spots (Tuesday, 6:30-7:30 am), and Mzee Otieno's matatu drivers logged 12 near-misses. (a) The long stretch had 140 of the 182 pedestrians. What percentage is this? What fraction of the road's 8,000 m does the long stretch (5,250 m) make up? (b) Mzee Otieno's log has 9 of 12 near-misses in the long stretch. What percentage is this? (c) Write a claim, evidence and reasoning paragraph that answers the driving question: How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story?	(a) $140 / 182 \times 100 = 77\%$ (to the nearest whole number). The long stretch is 5,250 m out of 8,000 m, which is about 65.6% of the road. (b) $9 / 12 \times 100 = 75\%$. (c) Claim: The officer's mean gap of 400 m on Thika Road is true but incomplete, because the road is not equally safe everywhere. Evidence: For the 20-gap sample, the median gap is 250 m and the mode is 200 m, and 14 of 20 gaps are 300 m or shorter. Six gaps of 600 m to 1,100 m (G9-G14) form a 5,250 m stretch that includes the school gate. In it we counted 140 of 182 undesignated crossers (77%) and Mzee Otieno's log has 9 of 12 near-misses (75%). The histogram and polygon show a cluster of short gaps and a separate group of very long gaps. Reasoning: A mean adds all gaps and shares them out equally, so a few very long gaps raise it and hide where people actually cross. The median, mode, grouped table, histogram and polygon together show the spread and shape, and they point to the long stretch as the place where pedestrians are at risk. Our statistics let us check a claim using our own road data instead of relying on one number.

5. Recommendations and limits

(a) What two recommendations would you give the school principal, the Parents Association, the area chief and the county roads representative? Support each with data. (b) Give one reason why our findings should be treated carefully before we are fully certain. (c) Answer the KICD question: How do we use statistics in day-to-day life? Use an example from this project.	(a) First, add extra crossing points, such as a zebra crossing or footbridge near the school gate, because the long stretch G9-G14 has 77% of undesignated crossers (140 of 182) and 9 of 12 near-misses. Second, add crossing points in the gaps of 1000 m to <1200 m (3 gaps on Thika Road), since these are the longest distances with no designated crossing. A repeat count should also be done. (b) Our data are from one morning only (Tuesday, 6:30-7:30 am) on one road, so the count should be repeated on other days and times before we are certain. (c) We use statistics to check claims, make informed decisions and communicate with the community. In this project we collected data by observation (counting and measuring), organised it in tables, summarised it with mean, median and mode, and showed it in graphs. This showed that the officer's 400 m average was true but incomplete and helped us tell the community where action is needed.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Calculations and use of data	All calculations (mean, estimated mean, frequency density, percentages) are correct, show clear working and include units (m) and the 20-gap sample.	Most calculations are correct with some working shown; occasional missing units or small slips.	Several calculation errors or no working shown; units often missing.
Understanding of mean, median, mode and grouped data	Explains clearly why the mean of 400 m is true but incomplete, using median 250 m, mode 200 m, and why grouped results are estimates.	States that the mean is true but incomplete and compares with median or mode, with limited explanation of grouping.	Only restates the mean or confuses mean, median and mode.
Interpreting graphs and tables	Accurately describes the histogram, frequency density and polygons, and links shape to where the long gaps are and why similar means hide different shapes.	Describes the main features of the graphs but gives limited comparison of Thika Road and Mombasa Road.	Graph descriptions are incomplete or inaccurate; frequency density is not understood.
Claim, evidence and reasoning	Gives a clear claim supported by specific data (77%, 75%, G9-G14, 5,250 m) and reasoning that answers the driving question.	Gives a claim with some correct evidence, but the reasoning is incomplete.	Claim is unclear or lacks data evidence.
Recommendations and communication	Makes data-based recommendations, states the one-morning limitation and links statistics to daily life with correct	Makes reasonable recommendations with some data support; limited mention of limits.	Recommendations are not linked to data or are missing.

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